

Colin Sepulveda

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EDUCATION

Northwestern University, Weinberg College of Arts and Sciences

Evanston, IL

Bachelor of Arts in Computer Science, Bachelor of Arts in Economics

Expected June 2028

- Relevant Coursework: Computer Systems, Data Structures and Algorithms, Programming Languages, Econometrics, Linear Algebra, Multivariable Calculus

PROFESSIONAL EXPERIENCE

First Trust Portfolios L.P.

Wheaton, IL

Quantitative Technology Software Engineering Intern

June 2026 – Present

- Built and deployed a production GUI for a nightly financial-data validation pipeline, enabling operators to identify and snooze missing constituent-count checks across ~700 tracked indexes before automated on-call escalation
- Implemented structured logging for an automated system that coordinates rolling restarts and data reloads across 7 application servers, providing developers visibility into current and historical runs to diagnose reliability issues

PROJECTS

CSTCP (TCP-like protocol from scratch)

Evanston, IL

Project

April 2026 - Present

- Built a TCP-like reliable transport protocol in C++ using raw sockets, constructing Ethernet and IP headers from scratch for every packet rather than relying on the OS's built-in TCP/IP stack
- Implemented main TCP-style mechanics including a three-way handshake, connection teardown, sequence numbers, cumulative acknowledgments, retransmission timers, and a sliding window for flow control
- Tested the protocol on a physical two-node network built from 2 Raspberry Pi 5s connected through Ethernet, SSHing into each to run client/server instances, and used tcpdump/Wireshark to inspect and debug packet exchanges at the byte level

File Transfer over UDP

Evanston, IL

Project

March 2026

- Built a UDP-based file transfer system in C++20 with separate client/server executables, POSIX socket abstractions, session state machines, and binary file reconstruction for server-hosted assets
- Designed a custom packet protocol with typed messages, fixed-size headers, sequence numbers, checksums, and file fragmentation/reassembly logic that chunks binary files into ordered UDP datagrams for client-side reassembly
- Implemented asynchronous network I/O using a background receiver thread, poll-based socket readiness, and a custom thread-safe blocking queue using mutexes and condition variables

Kalshi Data API

Evanston, IL

Project

September 2025

- Developed a Python system for retrieving and analyzing Kalshi market data using REST and WebSocket APIs
- Built an asynchronous data pipeline with asyncio for concurrent market data processing
- Analyzed order book imbalance and microprice dynamics to study short-term price movement

ADDITIONAL INFORMATION

Certifications and Exams: FINRA Securities Industry Essentials (SIE) Exam – Passed (September 2025), Bloomberg Market Concepts, Bloomberg Finance Fundamentals

Languages: C++23, Python, SQL, Object Pascal

Systems & Networking: Linux, TCP/IP, UDP, POSIX Sockets, Raw Sockets, Ethernet, IPv4

Tools & Technologies: Git, GDB, Wireshark, tcpdump, Docker, WSL

Interests: Computer Architecture, Operating Systems, Investing, Chess, Soccer, Weightlifting, Guitar